

Unit-4

Electronic

Hardware

1. * Definitions.

1. Define IC. Give two examples.

Answer. IC (Integrated Circuit):-

When number of diodes, transistors and capacitors are permanently fabricated on a semi-conductor wafer then it forms a integrated circuit.

- Example :- Wrist watch, radio, computer.

2. Define Fan IN.

Answer. Fan IN:-

It is the maximum number of inputs which can be connected to a gate without degrading the voltage level.

3. Define Fan Out.

Answer. Fan Out:-

It is the maximum number of similar logic gates which can be driven by logic gate without degrading the voltage level.

4. Propagation delay.

Answer. Propagation delay.

The difference in time between input and output signal is known as propagation delay.

* Long Question:-

1. Explain PCB & its type.

Answer. PCB (Printed Circuit Board):

A PCB is an electronic assembly on which the copper conductors are placed to create an electrical connections between components.

- PCB is also known as printed wiring board (PWB).
- In PCB, all components are connected without wires, all components are connected internally, so it will reduce the complexity of the overall circuit design.
- PCB is used to provide electricity and connectivity between the components by which it functions the way it was designed.

- Different types of PCB:-

- 1) Single layer PCB.
- 2) Double layer PCB.
- 3) Multi layer PCB.
- 4) Rigid PCB.
- 5) Flex PCB.
- 6) Rigid-Flex PCB.

1) Single layer PCB:-

- A single layer PCB is also known as single sided PCB.
- This type of PCB is simple and most used PCB because these PCBs are easy to design and manufacture.
- One side of this PCB is coated with a layer of any conducting material. Generally, copper is used as a conducting material for PCB, because copper has very good conducting characteristic.

A layer of solder mask is used to protect PCB against oxidation followed by silk screen to mask out all of the components on the PCB.

- In this type of PCB, only one side of the PCB is used to connect different types of electrical or electronics components like resistor, capacitor, inductor, etc. These components are soldered.
- These PCBs are used in calculators, radio, printers and the solid-state drive.

2) Double layer PCB:-

- Double layer PCB is also known as double sided PCB.
- As name suggests, in this type of PCB, a thin layer of conducting material, like copper is applied to both top and bottom sides of the board.
- In PCB, on different layer of board, consist via, which has two pads in corresponding position on different layers.
- These are electrically connected by a hole through the board.
- This type of PCB is mostly used in industrial controls, converter, UPS system, HVAC application, phone, amplifier and power monitoring system.

3) Multi layer PCB:-

- Multi layer PCB has more than two layers.
- It means that, this type of PCB has at least three conductive layers of copper.
- For securing the board glue is sandwiched between that layer of insulation which ensures that the excess heat will not damage any component of circuit.
- This type PCB designing is very complex and used in very complicated and large electrical task in very low space and compact circuit.

- This type of PCB is used in large application like GPS technology, satellite system, medical equipment, file server and data storage.

4) Rigid PCB:-

- Rigid PCBs are made out of solid material which don't allow PCB from twisting.
- Same as flex PCB Rigid PCB also have different layer configuration like single layer, double layer and multilayer Rigid PCBs.
- Shape of this PCB is not change after installation.
- This PCB cannot be bent according to shape of base that's why this PCB is known as Rigid PCB.
- Lifespan of this type of PCB is very high, so this is used in many parts of computer like RAM, GPU and CPU.

5) Flex PCB:-

- Flexible PCB is also known as Flex circuit.
- This type of PCB used flexible plastic material like polyimide, PEEK (Polyether ether ketone) or transparent conductive polyester film.
- The circuit board is generally place in folded or twisted.
- This is very complex type of PCB and it also contains different range of layer like single sided flex circuit, double sided flex circuit and multisided flex circuit.
- Flex circuit is used in organic light emitting diode, LCD fabrication, flex solar cell, automotive industries, cellular telephones, camera and complex electronic devices like laptops.

6) Rigid-Flex PCB:-

- Combination of flexible circuit and rigid circuit is most important board.
- Flex-rigid boards consists of multiple layers of flexible PCB attached to a number of rigid PCB layer.
- It is used in cell phones, digital cameras and automobiles etc.

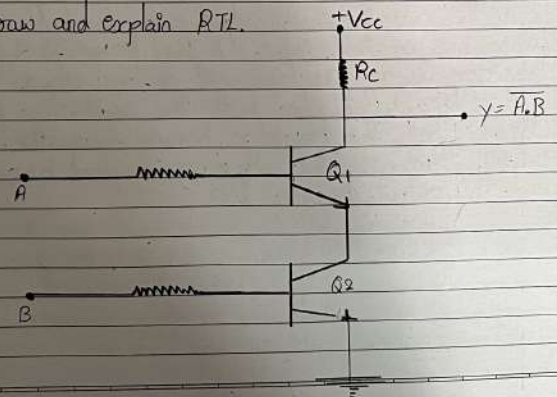
2. Compare SST, MSI, LSI, & VLSI

Answer. Below is a list of each of the generations the approximate capacity of each chip:-

- SST (Small-scale integration):- 1 to 10 transistors and 1 to 12 logic gates.
- MSI (Medium-scale integration):- 10 to 500 transistors and 13 to 99 logic gates.
- LSI (Large-scale integration):- 500 to 20,000 transistor and 100 to 9,999 logic gates.
- VLSI (Very-large-scale integration):- 20,000 to 1,000,000 transistors and 10,000 to 99,999 logic gates.

3. Draw and explain RTL.

Answer.



- There are two inputs given A and B.
- Two transistors are used.
- Three resistors are connected in a circuit.
- One source $+V_{cc}$ is given
- Output is y.

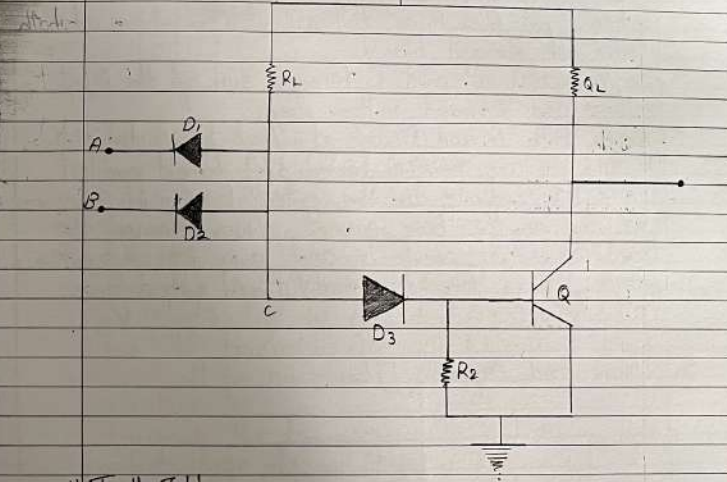
⇒ Truth Table:-

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

- When both A and B are at logic 0 both the transistors are off so no current flows through resistance R_c and the output is $+V_{cc}$ at logic level 1.
- When A or B or both are given logic 1 one or both the transistors are made on voltage is dropped across R_c as the current flows output goes to logic 0.
- When one or both inputs A and B are at logic 0 transistor Q_1 and Q_2 are off so the output becomes high (logic 1). But when both the inputs are high (logic 1) both the transistors become on so the collector current flows. so the voltage is dropped across R_c and the output becomes 0.

4. Draw and explain DTL.

Answer:



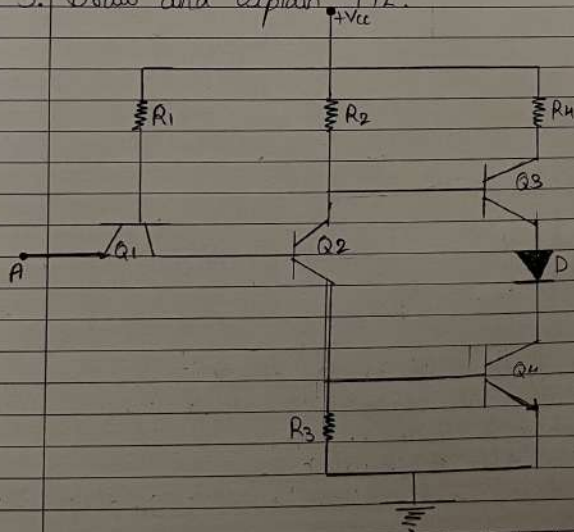
⇒ Truth Table:-

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

- There are three diodes in the circuit.
- Three resistors are connected in circuit.
- One transistor is used.
- One source $+V_{cc}$ is provided.
- Diode 1, diode 2 and resistor work as an AND gate.

- Transistor works as an inverter.
- This makes DTL NAND gate.
- When input A or B or both are at 0 level one or both diodes are forward biased.
- So potential of point C becomes sum of the input voltage and forward voltage drop.
- When both A and B are at logic 1, both the diodes D_1 and D_2 are reverse biased and do not conduct.
- Diode D_3 conducts as the potential of point C becomes $+V_{CC}$ and as the base current flows through the transistor Q_1 , it conducts and goes into saturation.
- Voltage V_{CC} is dropped across R_1 and output goes to logic 0.

5. Draw and explain TTL.

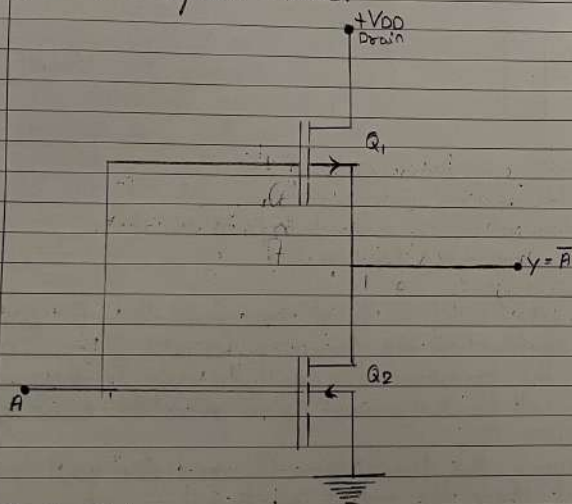


Truth Table:-

A	Y
0	1
1	0

- There is one input given A.
- Four transistors are used.
- Four resistors are connected in a circuit.
- One source $+V_{CC}$ is given.
- Output is Y.
- When the input A is at low level then the emitter-base junction of the transistor Q_1 will be forward biased. Thus, the transistor Q_1 will be ON. In this condition, there will be no current flow from the base of transistor Q_2 . It means the transistor Q_2 and Q_4 will not be conduct. Hence, the output is at high.
- When the input A is at high level, then the emitter-base junction of the transistor Q_1 will be reverse biased and the collector-base junction will be forward biased. Thus, the transistor Q_1 will be OFF. In this condition, the base current will be flow from the base of transistor Q_2 . It means the transistor Q_2 and Q_4 will be conduct. Hence, the output is goes to low.

6. Draw and explain CMOS.



Truth Table:-

A	Y
0	1
1	0

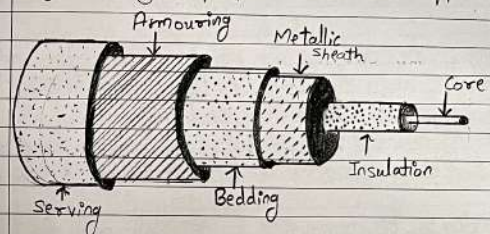
- Q₂ N channel is MOSFET which works as a input device.
- Q₁ P channel is MOSFET which works as a active load.
- When input A is on 0 level at that time Q₂ is OFF and Q₁ is ON. so output +VDD is logic 1.
- When input is logic 1 at that time Q₂ is ON and Q₁ is OFF so output is at logic 0 level so output is

7. Explain cable and connectors.

Answer

Cable:-

The cable is the group of one or more insulated conductors which are put together with number of layers to give proper mechanical support to them.



The general construction of the cable is shown in above figure. The main parts of the cable are described below:-

- 1) Core or Conductor:- The core is nothing but the conductor. Its main function is to carry the current. The core consists one or more than one conductors. The cable with single conductor is known as single core cable. Similarly, the cable with three conductors is known as three core cable. The conductors are made from the copper or aluminium. Generally, the stranded conductors are used which increases the flexibility of the cable.
- 2) Insulation:- Each core is covered by insulation layer of proper thickness. The thickness of the insulating layer depends on the operating voltage. The materials used for the insulating are paper, varnished cambric, vulcanized bitumen, polyvinyl chloride etc.

- 3) **Metallic sheath:-** The metallic sheath is provided over the insulating materials. Its main function is to protect the insulating material from the moisture, gases and other harmful liquids in the soil. They are generally made from lead, lead alloy or aluminium.
- 4) **Bedding:-** The bedding is provided over the metallic sheath. Its main function is to protect the metallic sheath from the corrosion and mechanical injury due to armouring. It consists of proper tape compounds with a fibrous material like jute or hessian tape.
- 5) **Armouring:-** The armouring is provided over the bedding. Its main function is to protect the cable from the mechanical injury during its laying and handling process. It consists of one or two layers of galvanised steel wires or two layers of steel tape.
- 6) **Seawing:-** The seawing is provided over the armouring. Its main function is to protect the armouring from the atmospheric conditions. It consists of fibrous material like jute cloth.

→ Connector:-

- **RJ45 connector:-**
- The RJ45 connector is a small, plastic cup which will be used to connect the wire inside the connector and ready to connect to the Internet.
- The RJ45 connector looks similar to a telephone jack but it looks slightly wider. The Ethernet cables are sometimes called RJ45 cables.

- In RJ45 the "RJ" stands for the Registered Jack and the "45" simply refers to the numbers of interface standards in the cable.
- They are easily installed.
- They are very reliable.
- Connection speed are higher.

→ BNC Connector:-

- BNC is also referred to as the Bayonet Nut Connector or British Naval Navy connector.
- A BNC connector is a type of connector used with coaxial Ethernet cable.
- The connector is bayonet-style meaning that it is put in then turned and locked in position.
- This connector is commonly used on a Token Ring network.
- Single design.
- Easy connection. They do not require any tools.

TRANSISTOR

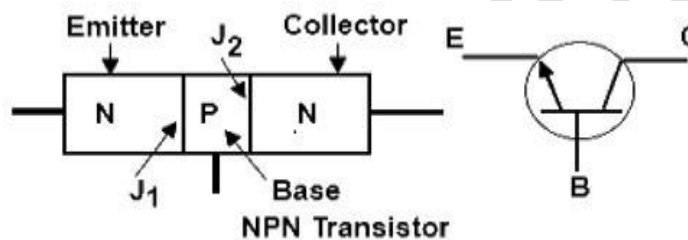
❖ Explain construction of transistor

There are two types of transistor

1. NPN Transistor
2. PNP Transistor

1. NPN Transistor

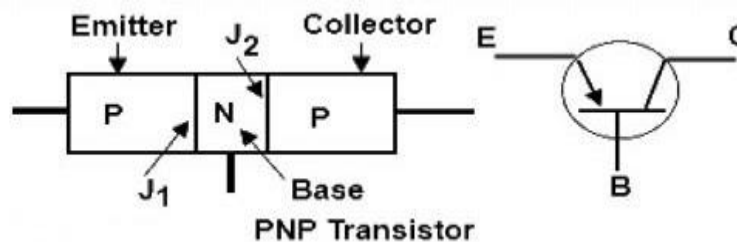
- Transistor has three layers named as NPN
- When the P- region is sandwiched between two N-regions, then the transistor is known as N-P-N transistor.
- There are three terminal named as a Base, Emitter and collector.



- Also symbol is shown in a diagram

2. PNP transistor

- This transistor has three layer named as PNP
- When the N- region is sandwiched between two P-region, then the transistor is known as P-N-P transistor.
- There are three terminal named as a Base, Emitter and collector.



- Also symbol is shown in a diagram

Emitter

- The region situated in one side of transistor, which supplies charge carriers to the other two regions.
- The emitter region is a heavily doped region

Base

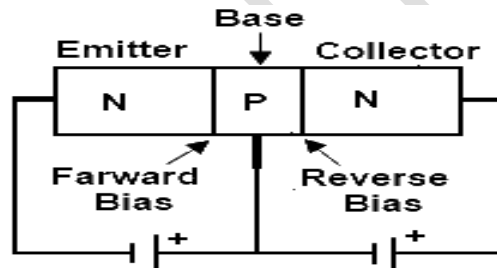
- It is the middle region that forms two P-N junctions in the transistor.
- The base of transistor is thin and lightly doped region.

Collector

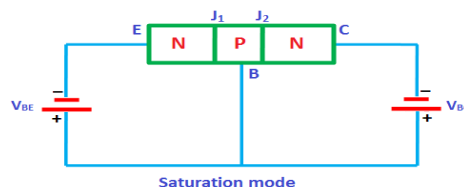
- It is a region situated in the other side of transistor, which collect the charge carriers.
- The collector of a transistor is always larger than the emitter and base if transistor.
- The doping level of the collector is intermediate between the heavy doping of emitter and the light doping of the base.

❖ Biasing of transistor

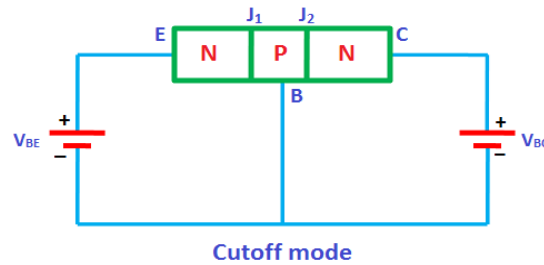
1. **Forward/Active** – In this mode, the emitter-base junction of a transistor is forward biased and the collector base junction is reverse bias is shown in figure. In a forward biasing the negative terminal of a battery is connected to N-side (Emitter) and positive terminal to P-side (Base).



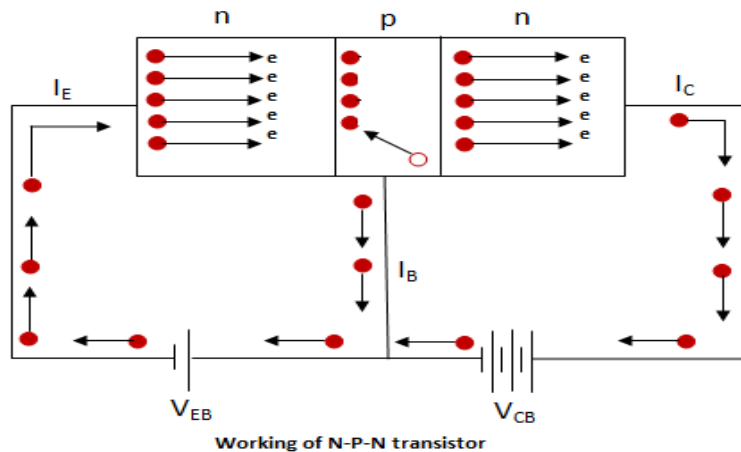
2. **Saturation**- In this mode, both the emitter-base and collector-base junctions of a transistors are forward-biased as shown in figure. In this mode transistor has a very large current. The transistor operated in this mode as a closed switch.



3. **Cut off mode**- In this mode, both emitter-base and collector-base junction of a transistor are reverse biased shown in figure. In this mode, the transistor has practically zero current. The transistor operated in this mode as an open switch.

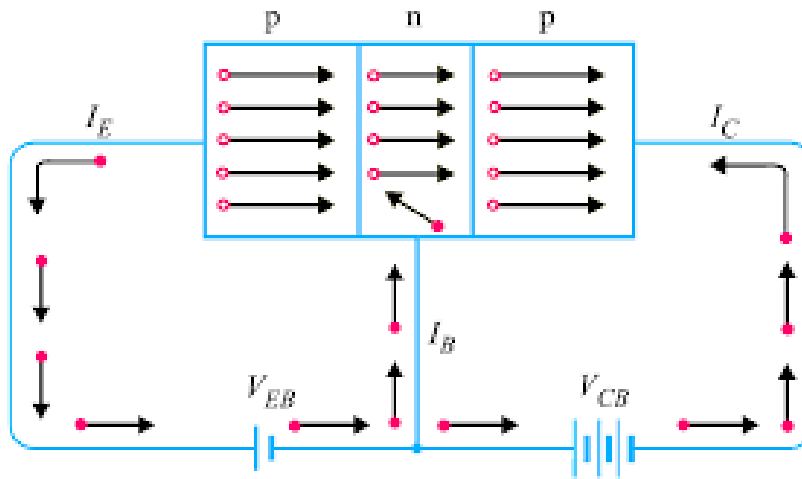


❖ Explain working of N-P-N transistor



1. NPN transistor is in forward bias, i.e. the base emitter of a transistor is forward biased and collector base junction is reverse biased. Is shown in figure.
2. The forward bias on the emitter-base junction causes the free electrons in the N-type emitter to flow towards the base region. This constitutes the emitter current (I_E).
3. Therefore, electrons, after reaching the base region tends to combine with the holes. They constitute the base current (I_B).
4. The free electrons of N-region are not combine with the P-region holes because of the lightly doped and small width of P-region.
5. Only 2 to 5 % electrons combine with the hole,
6. Remaining 95 to 98 % of electrons are diffused in the N- region (collector). They constitute the collector current (I_C).

❖ Explain working on P-N-P transistor

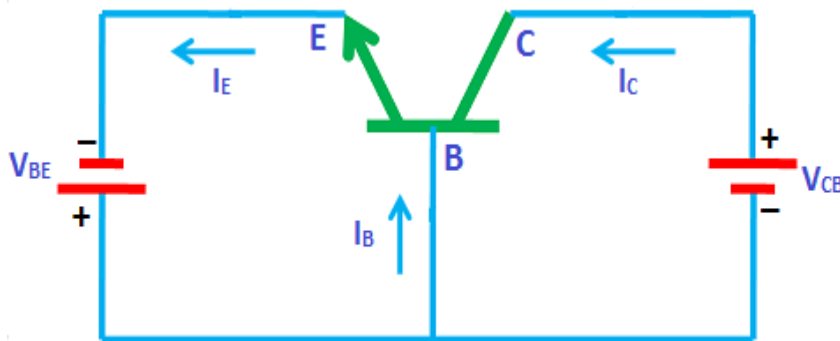


Working of P-N-P transistor

1. PNP transistor is in forward bias. i.e. the base emitter of a transistor is forward biased and collector base junction is reverse biased. Is shown in figure.
2. The forward bias on the emitter-base junction causes the holes in the N-type emitter to flow towards the base region. This constitutes the emitter current (I_E).
3. Therefore, holes, after reaching the base region tends to combine with the electrons. They constitute the base current (I_B).
4. The holes of P-region are not combine with the N-region electrons because of the lightly doped and small width of N-region.
5. Only 2 to 5 % electrons combine with the electrons,
6. Remaining 95 to 98 % of electrons are diffused in the P- region (collector). They constitute the collector current (I_C).

❖ **Explain Transistor configuration.**

1. Common Base configuration
2. Common Emitter configuration
3. Common Collector configuration

1.Common Base configuration**Diagram****1. Common bas D.C current gain**

D.C current gain is defined as the ratio of Transistor output collector current to the Transistor input emitter current.

It is denoted by 'α'

$$\alpha = \frac{\text{Transistor output Collector current } (I_C)}{\text{Transistor input Emitter current } (I_E)}$$

$$\alpha_{dc} = \frac{I_C}{I_E}$$

- In Transistor the collector current is always less than emitter current.
- Therefore current gain of common base configuration is always less than unity.
- The value of D.C current gain is closer to unity by making the width and the doping level of base region as small as possible.
- This causes the majority of electrons in the NPN transistor to reach the collector. Hence the collector current is higher.
- The actual value of α ranges from 0.95 to 0.99 %

We will find the collector current

$$I_C = \alpha \cdot I_E$$

We also know that the emitter current

$$I_E = I_B + I_C$$

$$I_B = I_E - I_C$$

Put the value of I_C in above equation

$$I_B = I_E - \alpha \cdot I_E$$

$$I_B = (1 - \alpha)I_E$$

2. Common base A.C current gain

It is defined as the ratio of small change in **collector current** (ΔI_C) to a small change in **emitter current** for constant collector to base (ΔI_B) voltage.

$$\alpha_{ac} = \frac{\Delta I_C}{\Delta I_E}$$

Where, ΔI_C = change in collector current

ΔI_E = change in emitter current

Now,

$$I_E = I_C + I_B$$

$$\Delta I_E = \Delta I_C + \Delta I_B$$

Dividing by ΔI_E on both side

$$\frac{\Delta I_E}{\Delta I_E} = \frac{\Delta I_C}{\Delta I_E} + \frac{\Delta I_B}{\Delta I_E}$$

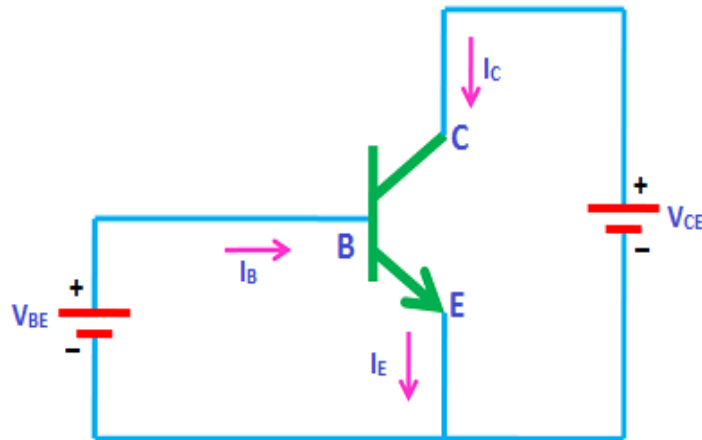
$$1 = \alpha_{ac} + \frac{\Delta I_B}{\Delta I_E}$$

$$\alpha_{ac} = 1 - \frac{\Delta I_B}{\Delta I_E}$$

- It clearly shows that the value of current gain is less than unity. ($\alpha_{ac} < 1$)
- The value of α_{ac} approaches to unity if the value of base current (I_B) reduces to zero.
- This can be achieved by doping the base lightly and making it very thin.
- The practical value of α_{ac} in commercial transistor is 0.95 to 0.99
- It is due to the fact that the output resistance of a common base transistor is much higher than the input resistance.

2. Common emitter configuration

Diagram



1. Common emitter D.C current gain.

Common emitter D.C current gain is defined as the ratio of **Transistor output collector current** (I_C) to the **Transistor input base current** (I_B) .

$$\beta_{dc} = \frac{\text{Transistor output Collector current } (I_C)}{\text{Transistor input Emitter current } (I_E)}$$

$$\beta_{dc} = \frac{I_C}{I_B}$$

- We know that collector current of a transistor is much larger than the base current.
- Therefore, the value of β is much greater than the unity. ($\beta_{ac} > 1$)
- Sometimes β_{dc} of transistor is also known as large signal common emitter current gain.

2. Common emitter A.C current gain

It is defined as the ratio of small change in **collector current** (ΔI_C) to the small change in **base current** (ΔI_B) for a constant collector to emitter voltage (V_{EB}).

It is denoted by β_{ac}

$$\beta_{ac} = \frac{\Delta I_C}{\Delta I_B}$$

❖ Explain Relation between current gain α and β .

We know that emitter current (I_E) of transistor is the sum of its base current (I_B) and collector current (I_C).

$$I_E = I_B + I_C$$

Dividing the above equation on both side by I_C

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + \frac{I_C}{I_C}$$

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + 1$$

$$\text{Since } \frac{I_C}{I_B} = \alpha \text{ and } \frac{I_C}{I_B} = \beta$$

$$\text{Therefore, } \frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\frac{1}{\alpha} = \frac{\beta+1}{\beta}$$

$$\alpha = \frac{\beta}{\beta+1}$$

Above equation may be written as

$$\alpha(\beta + 1) = \beta$$

$$\alpha\beta + \alpha = \beta$$

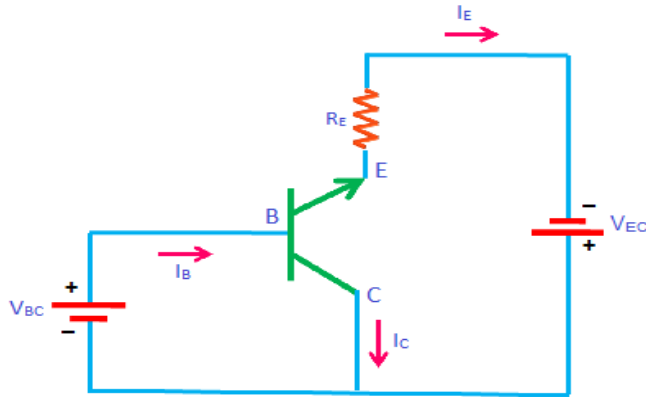
$$\alpha = \beta - \alpha\beta$$

$$\alpha = (1 - \alpha)\beta$$

$$\beta = \frac{\alpha}{1-\alpha}$$

3. Common collector configuration

Diagram



It is defined as the ratio of change in **emitter current** (ΔI_E) to the change in **base current** ΔI_B

It is denoted by γ

$$\gamma = \frac{\Delta I_E}{\Delta I_B}$$

❖ Explain Relation between current gain α and γ .

$$\gamma = \frac{\Delta I_E}{\Delta I_B} \dots\dots\dots (1)$$

$$\alpha = \frac{\Delta I_C}{\Delta I_E}$$

$$I_E = I_B + I_C$$

$$\Delta I_E = \Delta I_B + \Delta I_C$$

$$\Delta I_B = \Delta I_E - \Delta I_C$$

Substituting the value of I_B in equation (1), we get

$$\gamma = \frac{\Delta I_E}{\Delta I_E - \Delta I_C}$$

Dividing the above equation by I_E

$$\gamma = \frac{1}{1 - \frac{\Delta I_C}{\Delta I_E}}$$

$$\gamma = \frac{1}{1 - \alpha} \dots\dots\dots (2)$$

Put the value of $\alpha = \frac{\beta}{\beta + 1}$ in equation 2 we get

$$\gamma = \frac{1}{1 - \frac{\beta}{\beta + 1}}$$

$$\gamma = \beta + 1$$

❖ Transistor characteristic

❖ Explain characteristic of a transistor in a common base configuration

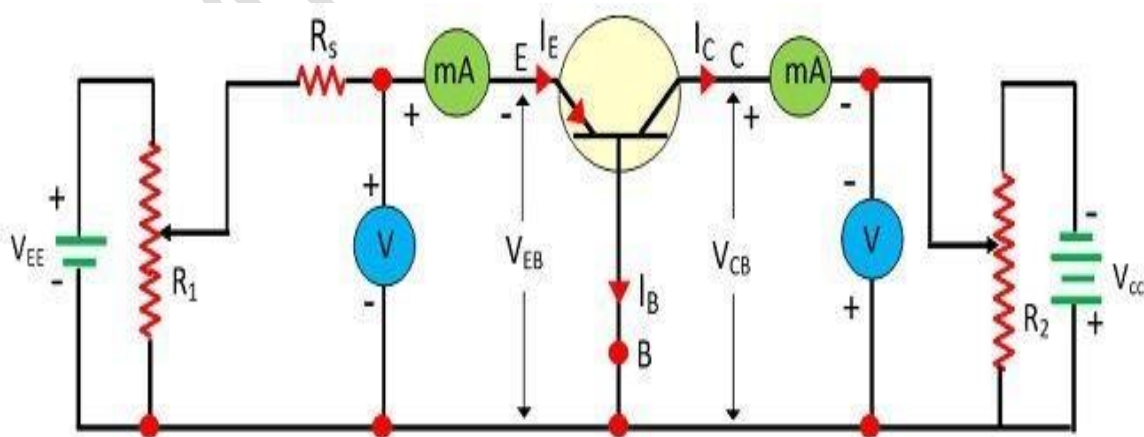
There are two type of characteristic

1. Input characteristic
2. Output characteristic

Input characteristic

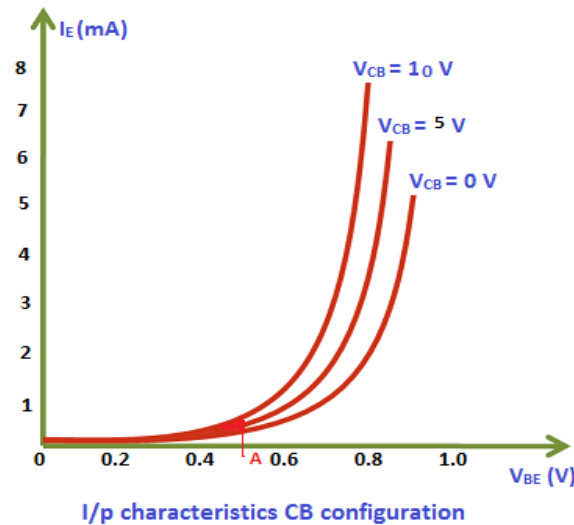
- Input characteristic gives the relation between emitter current and (I_E) and the emitter to base voltage (V_{EB})

Circuit diagram



Characteristic

- Input characteristic gives the relation between emitter current and (I_E) and the emitter to base voltage (V_{EB}). Is shown in a diagram.

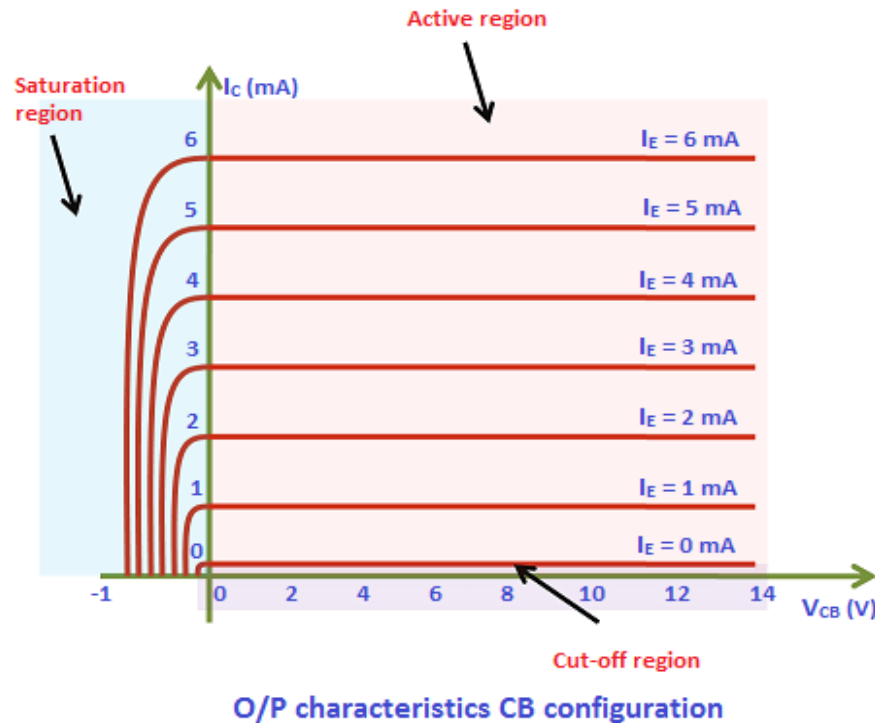


- (V_{EB}) on X- axis and (I_E) on Y- axis.
- First of all, adjust the collector base voltage at 1v
- Then increase the emitter to base voltage in small suitable step (i.e. 1v) and note down the emitter current
- There exists a threshold voltage also called offset voltage, cut in voltage or knee voltage as indicated by region OA.
- The value of knee voltage is 0.5 V for silicon and 0.1 V for germanium.
- Beyond point A, for a fixed collector to base voltage (V_{CB}), the emitter current (I_E) increase rapidly with small increase in emitter to base voltage (V_{EB}).
- It means that input resistance (R_i) of transistor in common base configuration is very small.

$$R_i = \frac{\Delta V_{EB}}{\Delta I_E}$$

Output characteristic

- Collector to base voltage on X-axis (V_{CB}) and collector current (I_C) on Y-axis.

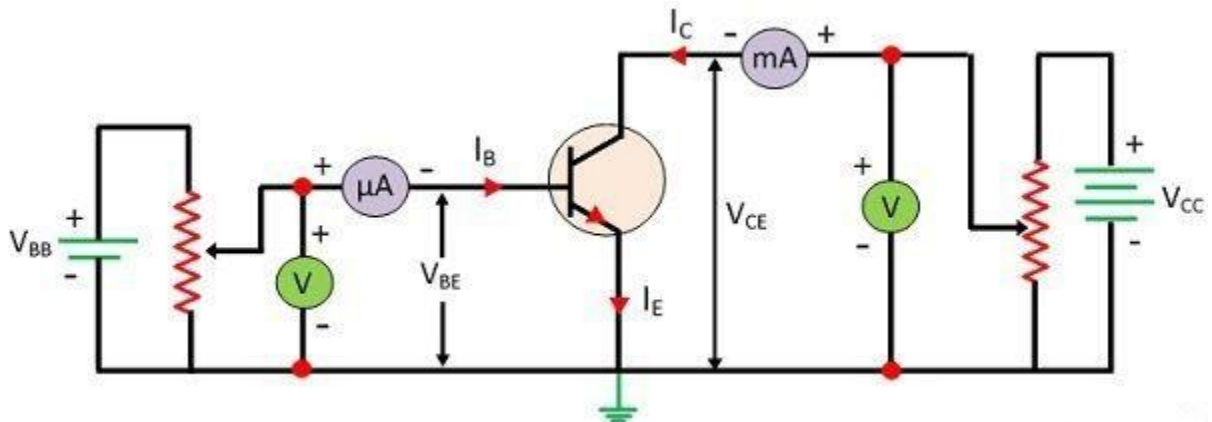


- First of all, adjust the emitter to base voltage (V_{EB}) to get suitable value of emitter current (I_E).
- Keeping the emitter current constant and increase the collector to base voltage (V_{CB}) from zero in a number of suitable step and record the corresponding value of collector current (I_C).
- The divided into three important regions namely saturation region, active region and cut off region.
- The saturation region is the region to the left of the vertical dashed line. it may be noted that in this region, collector to base voltage (V_{CB}) is negative for NPN transistor.
- It means that the collector base junction of a transistor is also forward bias in the saturation region.
- In this region small change in (V_{CB}) results in a large value of current.
- The active region is the region between the vertical dashed line and the horizontal axis.
- The characteristic may be used to find the value of common base transistor a.c output resistance.

$$R_o = \frac{\Delta V_{CB}}{\Delta I_C}$$

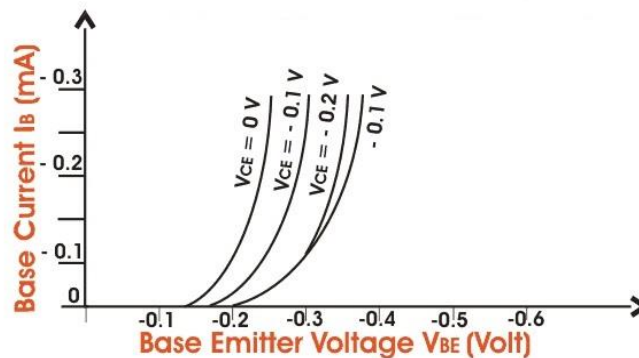
❖ Explain characteristic of a transistor in a common Emitter configuration

Circuit diagram



Input characteristic

- Increase base to emitter voltage (V_{BE}) in small suitable step and record the value of base current
- Graph shows the relation between (V_{BE}) which on X-axis and base current (I_B) which on Y-axis.

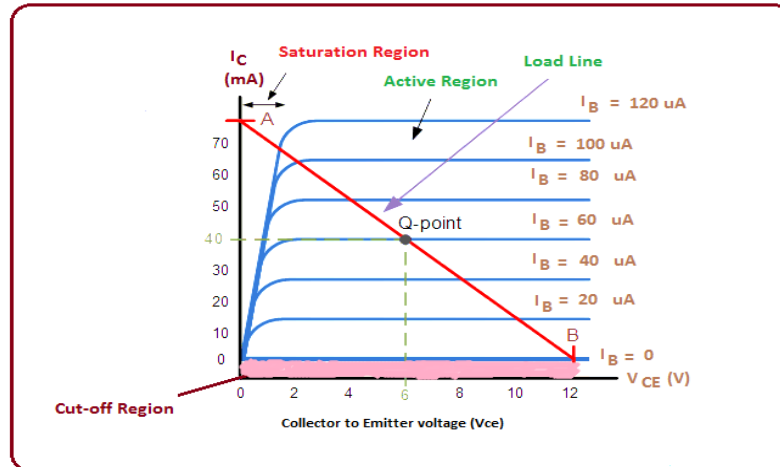


- There exists a threshold voltage or knee voltage below which the base current is very small. The value of knee voltage is 0.5 V and 0.1 V for germanium.
- After the knee voltage the base current (I_B) increase with the increase in base to emitter voltage for a constant collector to emitter voltage.
- it means that input characteristic of a resistance in common emitter configuration is higher as compared to the common base configuration.
- The input characteristic may be used to find out the value of common emitter transistor a.c input resistance.

$$R_i = \frac{\Delta V_{BE}}{\Delta I_B}$$

Output characteristic

- Adjust the base current (I_B) to 40 μA value.
- Then increase the collector to emitter voltage (V_{CE}) in a number of step and record the corresponding values of collector current (I_C)

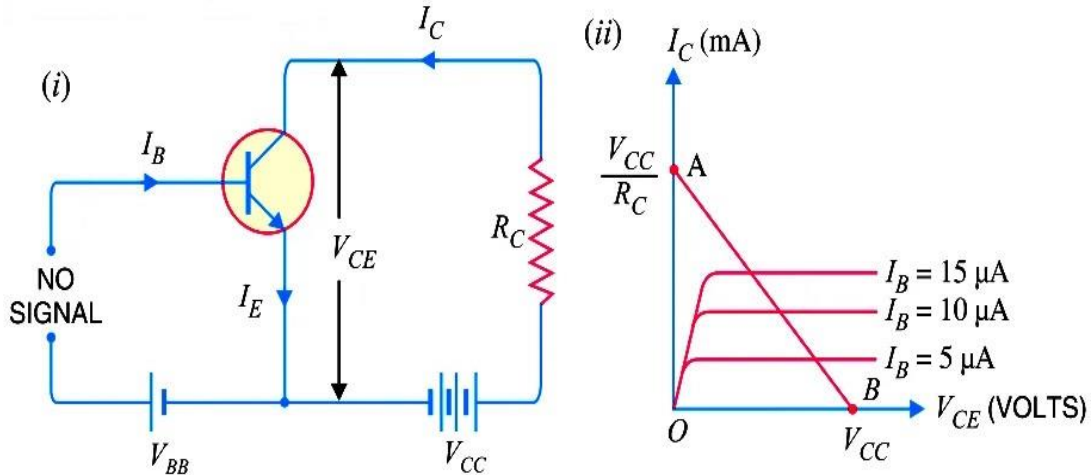


- Graph shows the relation between collector to emitter voltage (V_{CE}) which on X-axis and collector current (I_C) on Y-axis.
- The output characteristic divided in to three regions namely saturation region, active region and cut off region.
- The saturation and cut off regions are shown by the shaded areas.
- While the active region is the region between the saturation and cut off regions.
- The collector to emitter voltage (V_{CE}) is increased above zero, the collector current (I_C) increase rapidly to a saturation value.
- When the collector to emitter voltage (V_{CE}) is increased further the collector current (I_C) slightly increases.
- When the base current is zero, a small collector current exists. This is called leakage current.
- The output characteristic may be used to find out the common emitter transistor a.c output resistance.

$$R_o = \frac{\Delta V_{CE}}{\Delta I_C}$$

❖ What is D.C load line?

- Consider a common emitter NPN transistor circuit is shown in diagram



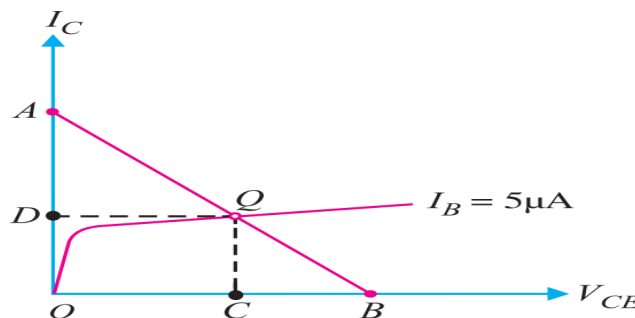
- Where no signal is applied.
- The value of collector emitter voltage V_{CE} at any time is given by

$$V_{CE} = V_{CC} - I_C R_C$$

- As V_{CC} and R_C are fixed value
- Therefore, it is a first degree equation can be represented by straight line on the output characteristic.
- This straight line (A & B) is known as **D.C load line**.

❖ What is operating point?

- The zero signal value of I_C and V_{CE} are known as the **operating point**.
- It is also called **Quiescent (silent) or Q – point**. Because it is the point on I_C and V_{CE} when transistor in the absence signal
- It is shown in below diagram.

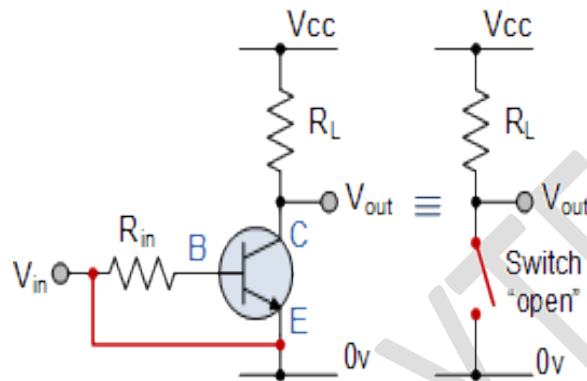


❖ Explain transistor as a switch

- Transistor has two switching state
 1. OFF state
 2. ON state

1. OFF state

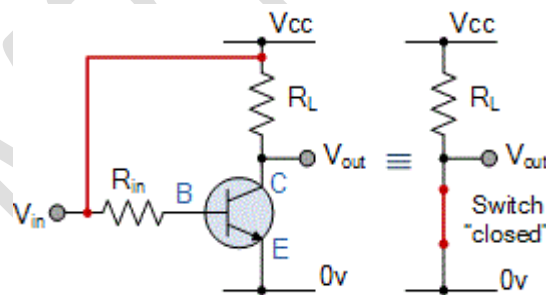
- In OFF state transistor operate in cut off region
- The diagram for the OFF state which shown below.



- When the input of transistor is zero or negative, the transistor is in OFF condition.
- In this condition, $I_B = 0$ and collector current is equal to the collector leakage current.

2. ON state

- In ON state transistor operate in saturation region.
- The diagram for the ON state which shown below.



- When the input voltage made positive that saturation collector current flows, the transistor is said to be ON condition.

❖ Testing of transistor

- There are three terminals is Base, Emitter and collector.
- There are three pair of terminal: B-E, B-C and C-E.
- To test the transistor, each pair of the terminal should be tested in forward bias and reverse biased.
- It means there are six test to be done to the transistor.
- The good transistor gives following result.
 1. **Emitter-Base (E-B)**-junction should behave as a diode and conduct only in one way.
 2. **Collector-Base (C-B)**- junction should behave as a diode and conduct only in one way.
 3. **Collector- Emitter (C-E)**- should not conduct either way.

❖ Short question**1. Advantages of transistor**

- Very small size and weight.
- Low operating voltage.
- Low cost.
- Low power consumption.
- Long life.

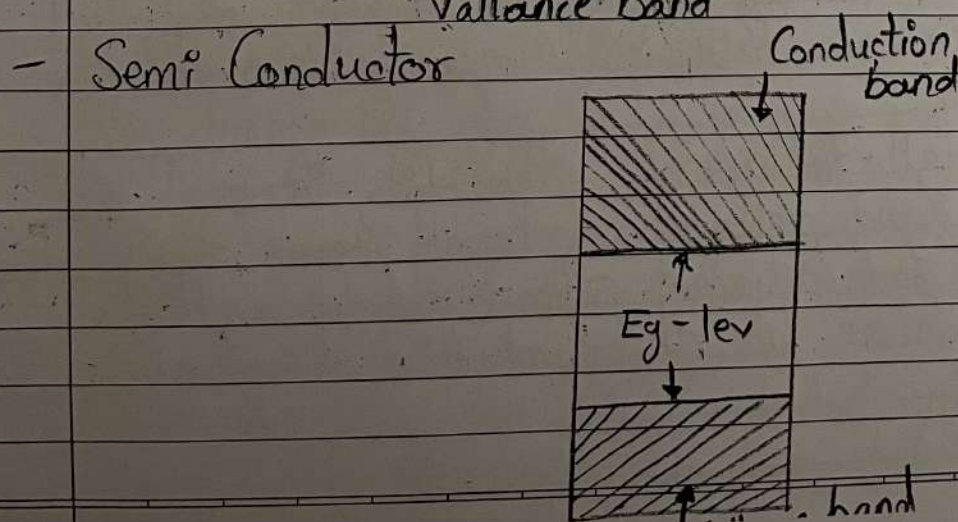
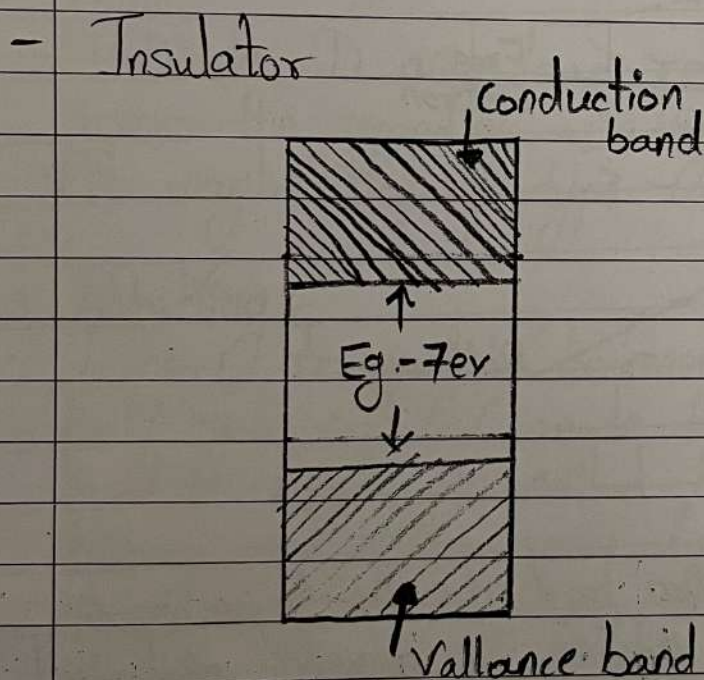
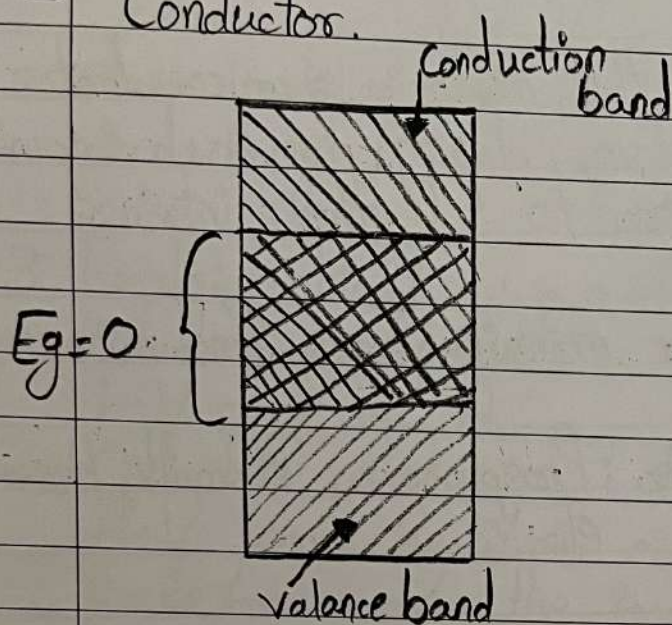
2. Disadvantage of transistor

- Lower temperature rating.
- Limited low voltage rating.
- Limited to low frequency operation.
- More sensitive.

3. Application of transistor.

- As a linear amplifier.
- As an electronic switch.

Q-1. Draw Energy band Diagram of Conductor, Insulator, and Semiconductor.



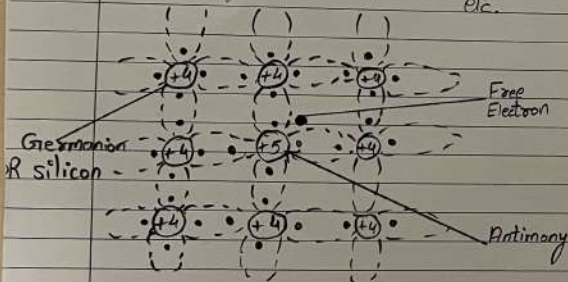
Q-2. Explain N-type semi Conductor.

Answer. - The five valence electrons impurity materials is known as donor impurity.

- When it is added to the intrinsic semiconductor materials. It is known as donor doping as each impurity atom donates one free electron to the intrinsic materials.

- These extrinsic semiconductor materials are known as N-type semiconductor.

- Example of Donor Impurities: Phosphorous, Bismuth, Arsenic etc.



- Explanation.

- Figure shows the formation of the N-type semiconductor material by adding the arsenic material into silicon. The arsenic material (impurity) has the five valence electrons.

- In silicon crystal the arsenic atom fits in such a way that its four valence electrons forms the covalent band with four adjacent silicon atoms and the fifth electron enters into the conduction band as a free electron.

- It means that each arsenic atom gives one free electron when it is added into the silicon atom. As the free electron has the negative charge such materials are known as the N-type semi conductors materials.

- Such number of free electrons can be controlled by the amount of the impurity added to the silicon.

Q-3. Explain Half wave Rectifier.

Answer. In Half wave Rectifier current flows only during the half cycle of A.C.

- Circuit:

Half wave Rectifier circuit is shown in figure and the waveforms of the input and output are shown in figure.

- Diode D and load resistance are connected in series across the secondary winding of a step down transformer. (It may be step up type also).

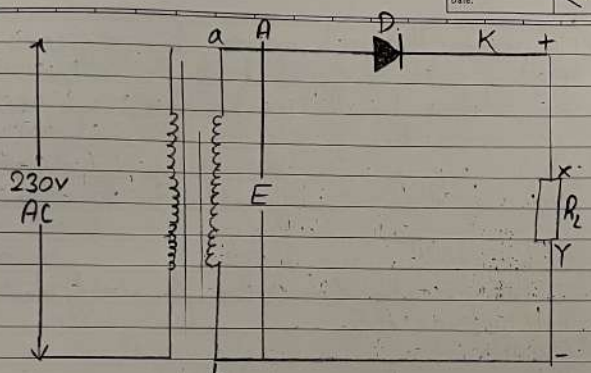
• Working:

- When terminal 'a' becomes positive with respect to 'b' in the positive half cycle of A.C. anode A of the diode D becomes positive with respect to its cathode K. The diode is forward biased and offers very low resistance.

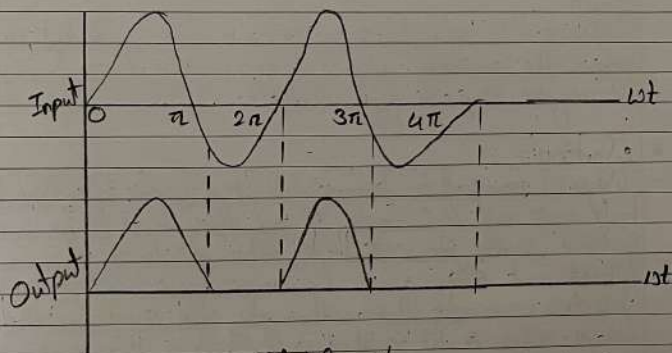
- Electric current flows through the load in the direction a → A → K → Y → b → a. That is the current flows X to Y through the load.

- But during the negative half cycle when terminal 'a' becomes negative with respect to 'b', the diode is reverse biased and offers very high resistance and current does not flow. Waveforms of the input voltage and output current are shown in figure.

- It is seen from the waveforms that current conducts only in one direction. That is current flows during half cycle of a.c. so it is called the half wave Rectifier.



(a) Circuit

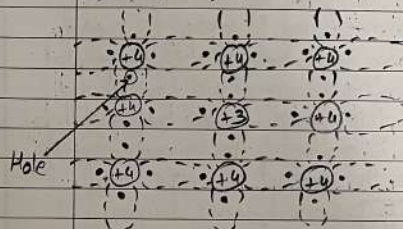


(b) Waveforms

4. Explain P-type semiconductor.
- Ans: The other type of impurity which has only three free valance electrons is known as trivalent atom such an impurity is known as acceptor impurity.
- When it is added to the intrinsic semiconductor materials, it is known as acceptor doping as it creates the

Avuva

- holes and they are ready to accept the electrons.
- These extrinsic semiconductor materials are known as P-type semiconductor.
- Example of Acceptor Impurities: Gallium, Boron, Indium etc.



• Explanation:

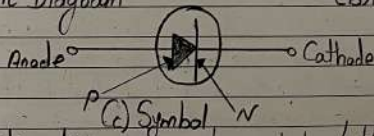
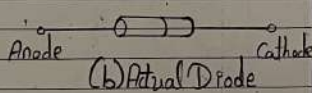
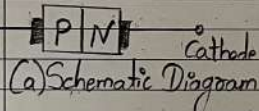
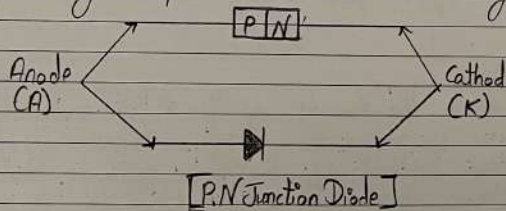
- Figure shows the formation of the P-type semiconductor material by adding the gallium material into silicon. The gallium material has the three valance electrons.
- In silicon crystal the gallium atom fits in such a way that its three valance electrons forms the covalent bond with three adjacent silicon atoms but due to the short of the fourth electron the fourth covalent band is not complete.
- The remaining is called a hole. It means that each gallium atom gives one hole when it is added into the silicon atom. As the hole has the positive charge such materials are known as the P-type semiconductor materials.
- Such number of holes can be controlled by the amount of the impurity added into the silicon.

* 7 marks:

1. Explain P-N Junctions, construction and working.
- The popular semiconductor device forming by P-N Junction is known as P-N Junction diode.

- Construction:

- The P-N Junction diode has two terminals one is from P-region and second is from N-region.
 - The P-region terminal is known as the anode.
 - The N-region terminal is known as the cathode.
 - A metal is applied to the P and N region to connect the external terminals such contacts between metal and P and N region are known as ohmic. Contact the important properties of this ohmic contact are given below.
1. In both the directions it conducts the current equally.
 2. The voltage drop across the contact is very small.



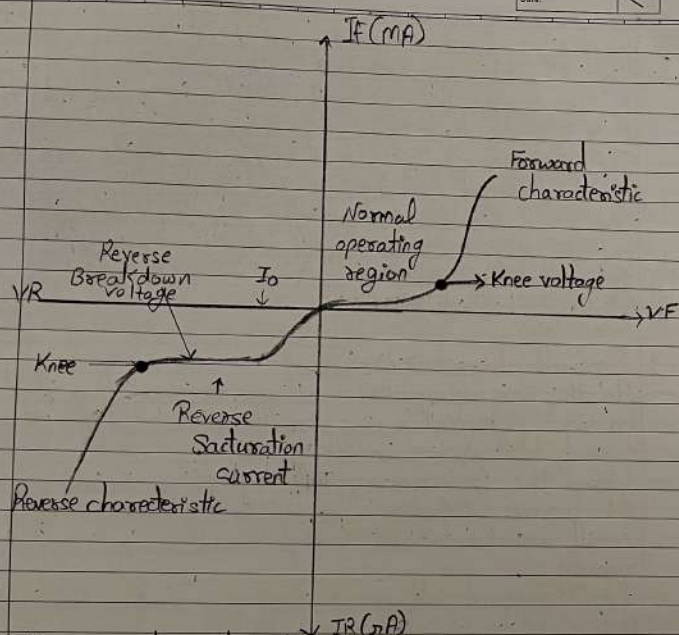
The schematic diagram and symbol of the P-N Junction diode is shown in figure respectively. In the diode the P-region acts as an anode while the N-region acts as a cathode. The arrow shows the conventional direction of the current when the external supply voltage connected to it.

Q-2 Explain V-I characteristic of P-N Junction diode.
 Answer - The V-I characteristic of the P-N Junction diode is the graph between the voltage applied to the P-N Junction diode and the current flowing through it. The performance of the P-N Junction diode can easily be judged by their V-I characteristic.

- The V-I characteristics of the P-N Junction diode contains two separate characteristics namely.
1. Forward characteristics.
 2. Reverse characteristics.

1. Forward characteristics:

- As the forward applied voltage is less than the barrier potential a very small amount of the forward current will flow from the diode.
- When the forward applied voltage increases close the barrier potential, the diode offers a very small resistance and the forward current increase rapidly.
- The forward voltage at which the forward current of the diode increases rapidly is known as the cut in voltage or knee voltage and the point. The value of the cut in voltage for silicon is 0.6 volt and for germanium is 0.2 volt.
- The forward characteristics and Reverse character of the diode is shown in figure.



2 Reverse characteristic:

- If the reverse applied voltage is less than the reverse breakdown voltage a very small amount of reverse saturation current will flow from the diode which is practically neglected.
- As the reverse applied voltage is increased beyond the reverse saturation current increases rapidly and it will be constant after a certain value of reverse voltage.
- The reverse applied voltage at which the reverse saturation current start to increase rapidly is known as reverse breakdown voltage. It is denoted by V_R .

Q-3. Explain full wave bridge Rectifier with input output wave forms.

Answer: In this type of rectifier there is no need to have centre tapping in the secondary winding of the transformer but four diodes are required in place of two.

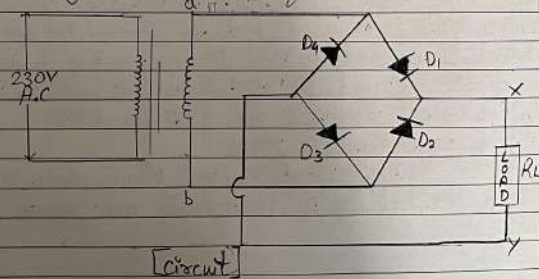
- The circuit configuration is similar to the wheatstone bridge so the circuit is called the bridge Rectifier.

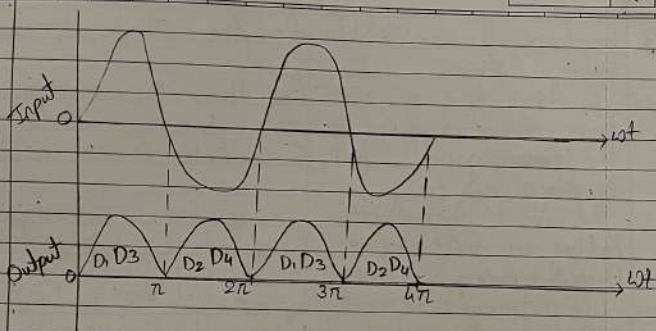
Circuit:-

It is a transformer cathodes of diodes D_1 and D_2 are joined together which becomes the Positive terminal of d.c.

- Anodes of diodes D_3 and D_4 are joined together which becomes the negative terminal of the D.C.

Anode of D_1 and cathode of D_4 are joined together and is connected to terminal 'a' of the transformer. Similarly cathode of D_3 and anode of D_2 are joined together and connected to the terminal 'b' of the transformer. Load is connected across the positive and negative terminals of the D.C output.





[Wave Forms]

Working:

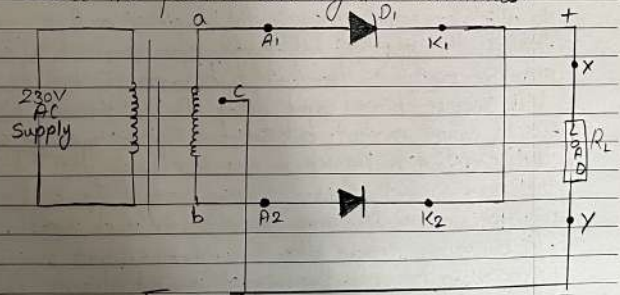
- During the positive half cycle of A.C when terminal 'a' becomes positive with respect to 'b' diodes D_1 and D_3 are forward biased and allow the current to conduct.
- At the same time diodes D_2 and D_4 are reverse biased and can not conduct.
- Current flows in the direction $a - D_1 - x - y - D_3 - b - a$
- During the negative half cycle terminal 'b' becomes positive with respect to 'a' diodes D_2 and D_4 are forward biased and D_1 and D_3 are reverse biased.
- So diodes D_2 and D_4 conduct and D_1 and D_3 can not. The current flows in the direction $b - D_2 - x - y - D_4 - a - b$ as shown in figure.
- It is seen that the load current flows from X to Y in both the half cycles waveforms are shown in figure.

Q-4. Explain Full wave rectifier with Input output wave forms.

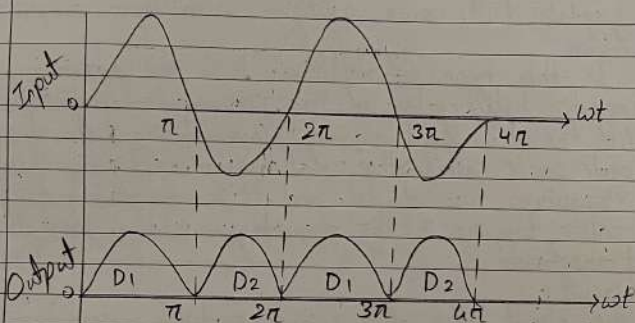
Answer. In this type of rectifier, both the positive and negative half cycles of a.c are converted into d.c i.e. the d.c load current flows during both the half cycle so it called the full wave rectifier. This requires two diodes and it is necessary to have center tapping in the secondary winding of the transformer.

Circuit:

- The circuit of a full wave rectifier using two diodes is shown in figure.
- Secondary winding of the transformer has a centre tapping C. D_1 and D_2 are two diodes whose cathodes are joined together. This is the positive terminal of the d.c.
- Anodes are connected to the two outer terminals of the transformer secondary centre tapping becomes the negative terminal of the d.c output. Load is connected across the positive and negative terminals.



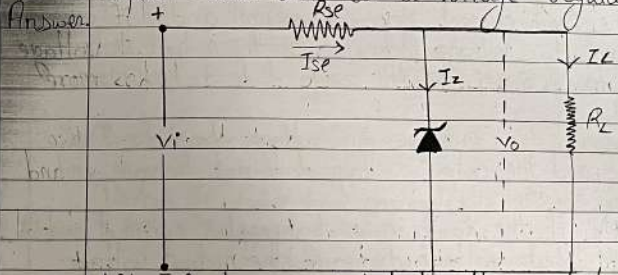
[Circuit]



Working

- When terminal 'a' becomes positive w.r.t 'c' during the positive half cycle of a.c., at the same time terminal 'b' becomes negative w.r.t 'c' so diode D_1 becomes forward biased and diode D_2 becomes reverse biased.
- So diode D_1 conducts and allows the current to conduct, where as diode D_2 does not allow the current to flow so electric current flow in the direction a A, K, x, y, c, a.
- During the negative half cycle when terminal 'a' becomes negative w.r.t 'c' terminal 'b' becomes positive diode D_1 is reverse biased and diode D_2 is forward biased.
- So D_1 cannot conduct but D_2 does Electric current flows in the direction b A, K, x, y, c, b. During this time also the load current flow from x to y. Terminal x becomes positive and y becomes negative of the d.c.
- In diagram shown the waveforms output waves do not enter in the negative direction and current flow during both the half cycle.

Q-5. Explain zener diode as a voltage regulator:

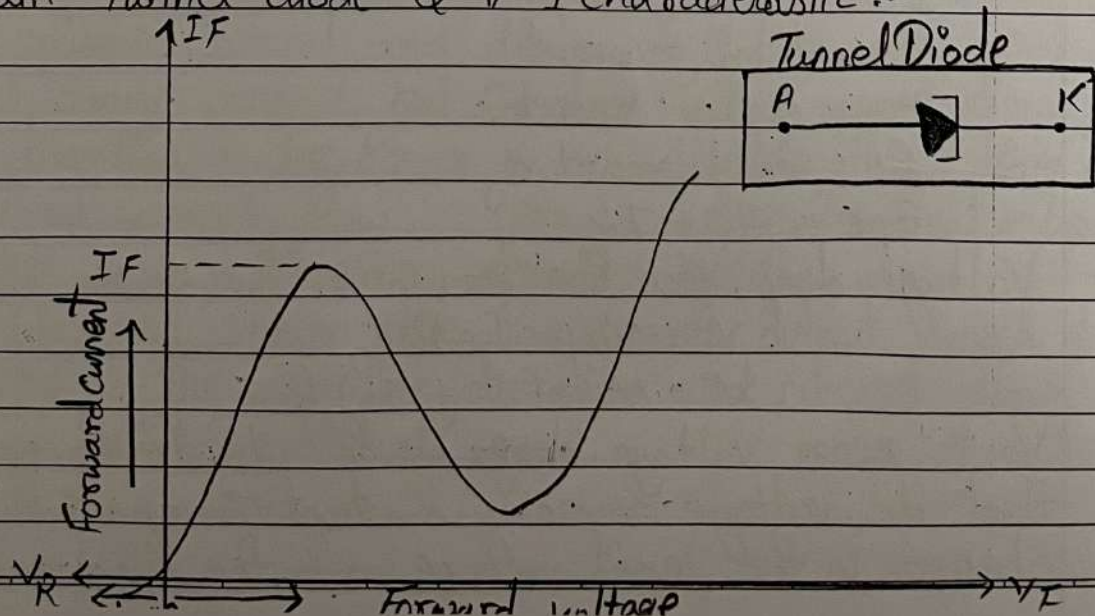


- When load is connected to the rectifier, the d.c load voltage changes with the change in the input d.c voltage applied to the rectifier or with the change in the load current or both. This d.c voltage across the load may be kept almost constant by making use of the zener diode.
- Zener diode with series resistor is connected in between the d.c output terminals of the rectifier and the load. The circuit is shown in diagram.
- Let V_i = input voltage
 V_o = output voltage
 I_L = load current
 I_z = Zener current
 I_{se} = total current
 R_{se} = series resistor
 R_L = load resistor
 $I_{se} = I_L + I_z$
- Voltage drop $I_{se} R_{se}$ occurs in the series resistor.
 $V_i = V_o + I_{se} R_{se}$
 $V_i = V_o + (I_L + I_z) R_{se}$
- V_o is zener voltage zener diode adjusts its current such that its voltage remains constant irrespective of the changes in the input voltage or in the load current.

- If the load current remains constant and if the input d.c voltage increases, so the output voltage tries to increase but the zener diode takes more current.
- So current I_{se} increase so voltage drop $I_{se} R_{se}$ increase and the output voltage does not increase and remains constant.
- Similarly if the load current remains constant and if the input voltage decreases. the output voltage tries to decrease but this time zener diode takes less current.
- So total current decreases and so series voltage drop decreases and output voltage V_o remains to the same value.
- Now let us try to understand its action when input voltage remains constant and the load current changes.
- If the load current increases with the constant input voltage, the total current tries to increase so the output voltage tries to decrease.

3-4 mark.

5. Explain Tunnel diode & V-I characteristic:-



BASIC OF ELECTRICAL ENGINEERING

❖ DEFINITION

1. Current

The flow of electric charge is known as an electrical current.

The unit of current is ampere

It is denoted by 'I'

2. Voltage or Potential Difference

The capacity of a charged body to do work its known as a voltage or potential difference.

The force required for the flow of electron is called voltage.

The unit of voltage is Volt.

It is denoted by 'V'

3. Power

The rate of doing work in an electrical field its known as power.

Electric power is a product of voltage (V) and current (I)

The unit of power is Watt

$$\text{Power (P)} = V \times I \quad \text{Watt}$$

$$1 \text{ HP} = 735.5 \text{ watt}$$

$$1 \text{ KW} = 1000 \text{ watt}$$

❖ Example:

A 100 watt, 250 voltage lamp has resistance of 250Ω , then calculate the current taken by the lamp also calculate the power.

Solution:

Given data

$$\text{Power (P)} = 100 \text{ watt}$$

$$\text{Voltage (V)} = 250 \text{ volt}$$

$$\text{Resistance (R)} = 250 \Omega$$

Calculation:**1. Current (I)****2. Power (P)****1. Current**

$$R = \frac{V}{I} \quad (\text{According to Ohm's law})$$

$$\therefore I = \frac{V}{R}$$

$$I = \frac{250}{250}$$

$$I = 1 \text{ Amp}$$

2. Power

$$\text{Power (P)} = V \times I$$

$$= 250 \times 1$$

$$= 250 \text{ Watt}$$

3. Resistance:

The property of material which oppose the flow of free electron, its known as resistance

The unit of resistance is Ohm

It is denoted by 'R'

4. Conductor

The property of material which allow the flow of electric current through in it. Its known Conductor. For example: Gold, Silver

5. Insulator

The property of material which resist the flow of electric current through in it. Its known Conductor. For example: Plastic, Rubber

6. Semi-conductor

A semi-conductor is a substance which has resistivity in between conductor and insulators

For example: Silicon, Germanium

7. Electrical energy

Electrical energy is the capacity to do work.

Electrical energy is also defined as the product of power and time.

The unit of electrical energy is joule or watt-second.

$$E = P \times t \text{ watt – second}$$

❖ Explain factor affecting to the resistance.

There are following factor affecting to the resistance.

1. Length of material
2. Cross sectional area
3. Types of material
4. Temperature

1. Length of material

- When the length of material increases, the resistance of the material is also increase.
- the length of material decreases, the resistance of the material is also decreases.
- It means length is directly proportional to resistance. $R \propto l$

2. Cross sectional area of material

- When the cross sectional area of material increase, the resistance of material is decreases.
- If the cross sectional area of material decreases, the resistance of material is increase.
- It means cross sectional area of material is inversely proportional to resistance $R \propto \frac{1}{a}$

3. Types of material

- Different types of material offer different value of resistance.
- The value of resistance is depending on the material

4. Temperature

- The resistance of material changes with changes in temperature.

❖ Explain Ohm's law with application and limitation.

Ohm's law established the relation between voltage and current.

If the temperature remains constant, the ration of voltage 'V' applied to conductor and current 'I' flowing through in it remain constant. It is identified as an Ohm's law.

$$\frac{V}{I} = \text{Constant}$$

$$\therefore R = \frac{V}{I} \quad (\text{R is constant})$$

Application:

- Unknown resistance 'R' can be found by using Ohm's law.
- Value of voltage drop across resistance in the circuit can be found using relation. $V = I \times R$
- Equivalent resistance of a circuit can be found.

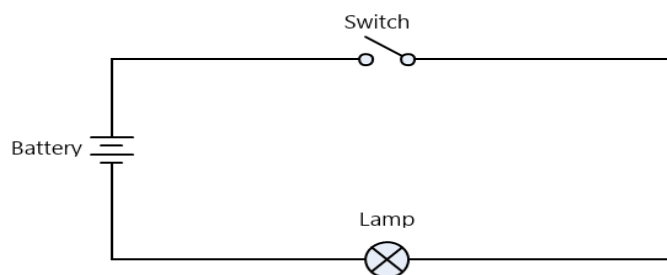
Limitation:

- Ohm's law can be applied when temperature remains constant.
- Ohm's law is not applicable to all material. For Ex. Semi conducting material.
- In AC circuit, Ohm's law can be applied to only resistive circuit.

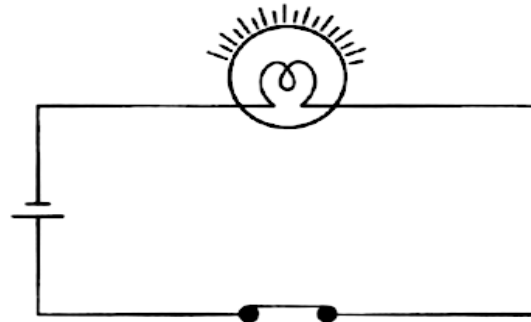
❖ Explain types of circuit

There are three types of circuit.

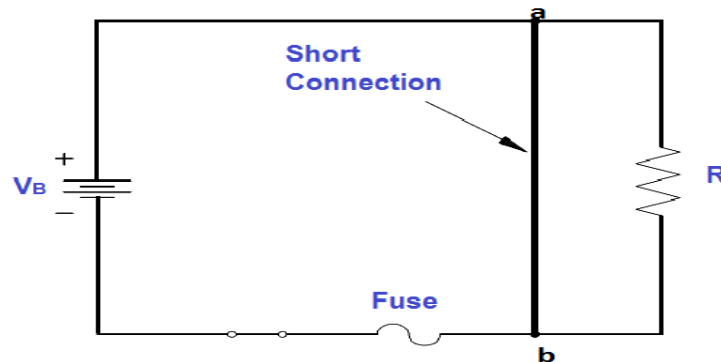
- Open circuit
- Closed circuit
- Short circuit

Open circuit**Diagram**

- As shown in a diagram a battery is used to provide supply voltage.
- A fuse is used for the protection of a circuit.
- A lamp and switch is also connected in a circuit.
- As shown in a diagram switch 'S' is open. Which is the break the path for the circuit.
- Such types of circuit are called Open circuit.

Close circuit.**Diagram**

- As shown in a diagram a battery is used to provide supply voltage.
- A fuse is used for the protection of a circuit.
- A lamp and switch is also connected in a circuit.
- As shown in a diagram switch 'S' is closed. Which is provide the continues path for the circuit.
- Such types of circuit are called Close circuit.

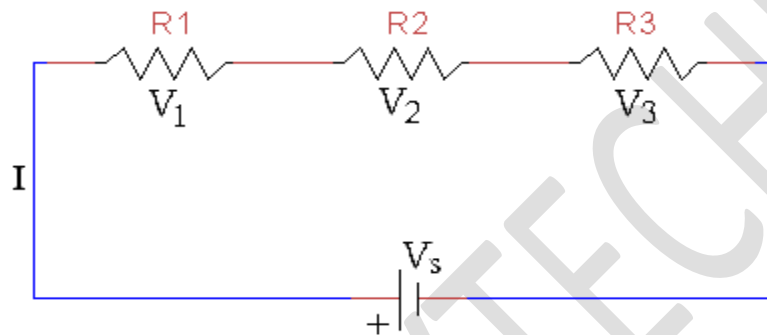
Short circuit**Diagram**

- Above figure shows the short circuit.
- Due to some reason point 'a' and 'b' are connected by a simple wire
- Now the resistance of this wire is very small compared to the load resistance.
- So large current flows through wire ab.
- This current is called short circuit I_s .
- As the fuse is kept in the circuit, it will blow off and the damage to the source is prevented.
- This type of circuit is called short circuit.

❖ **Draw and explain types of connection.**

There are two type of connection

- Series connection
- Parallel connection

Series connection**Diagram**

- In the series circuit the value of current is same.
- Due to these current there is voltage drop across each resistance.
- The voltage drop is proportional to the value of resistance.
- Sum of voltage is equal to the applied voltage.

$$V = V_1 + V_2 + V_3$$

$$IR = IR_1 + IR_2 + IR_3 \quad (\text{According to ohm's law } R = \frac{V}{I}, \quad \therefore V = IR)$$

$$IR = I(R_1 + R_2 + R_3)$$

$$\therefore R = R_1 + R_2 + R_3$$

And when more resistance is connected in series,

$$R = R_1 + R_2 + R_3 + R_4 + \dots$$

Power loss in the resistance

$$P_1 = I^2 R_1 \text{ Watt}$$

$$P_2 = I^2 R_2 \text{ Watt}$$

Example:

Three resistance of $10\ \Omega$, $20\ \Omega$ and $30\ \Omega$ are connected in series across 120 volts supply. Calculate 1). Equivalent resistance. 2) Current flowing through circuit. 3) Voltage drop across each resistance 4) Power drop in each resistance.

Solution**Given data**

$$R_1 = 10\ \Omega,$$

$$R_2 = 20\ \Omega,$$

$$R_3 = 30\ \Omega$$

$$\text{Voltage} = 120\ \text{V}$$

1. Equivalent resistance.

$$R_{eq} = R_1 + R_2 + R_3$$

$$= 10 + 20 + 30$$

$$R_{eq} = 60\ \Omega$$

2. Current flowing through circuit

$$R = \frac{V}{I} \quad (\text{According to Ohm's law})$$

$$I = \frac{V}{R}$$

$$I = \frac{120}{60}$$

$$I = 2\ \text{amp}$$

3. Voltage drop in each resistance

$$\text{Voltage drop in Resistance } R_1, V_1 = IR_1$$

$$V_1 = 2 \times 10$$

$$V_1 = 20\ \text{Volt}$$

Voltage drop in Resistance $R_1, V_2 = IR_2$

$$V_2 = 2 \times 20$$

$$V_2 = 40 \text{ Volt}$$

Voltage drop in Resistance $R_1, V_3 = IR_3$

$$V_2 = 2 \times 30$$

$$V_3 = 60 \text{ Volt}$$

4. Power loss in each resistance

Power loss in resistance $R_1, P_1 = I^2 R_1$

$$P_1 = 2^2 \times 10$$

$$P_1 = 4 \times 10$$

$$P_1 = 40 \text{ Watt}$$

Power loss in resistance $R_2, P_2 = I^2 R_2$

$$P_2 = 2^2 \times 20$$

$$P_2 = 4 \times 20$$

$$P_2 = 80 \text{ Watt}$$

Power loss in resistance $R_3, P_3 = I^2 R_3$

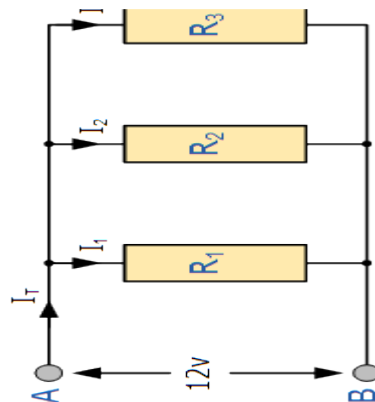
$$P_3 = 2^2 \times 30$$

$$P_3 = 4 \times 30$$

$$P_3 = 120 \text{ Watt}$$

Parallel connection

Diagram



- The parallel connection is shown in the diagram
- One lead of each resistance are connected together and similarly other lead are connected together.
- Voltage 'V' applied across these two terminal. So each resistance gets same voltage.
- The value of current 'I' is different and inversely proportional to each resistance.

$$I = I_1 + I_2 + I_3$$

$$\therefore \frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\therefore \frac{V}{R} = V \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$

$$\therefore \frac{V}{R} = V \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$

$$\therefore \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Power loss in the resistance

$$P_1 = I^2 R_1 \text{ Watt}$$

$$P_2 = I^2 R_2 \text{ Watt}$$

$$P_3 = I^2 R_3 \text{ Watt}$$

Example:

Three resistance of 20 Ω , 30 Ω and 50 Ω are connected in a parallel across 60 volts supply calculate the following

1. Current flowing through each resistance.
2. Total power.
3. Equivalent resistance.
4. Power loss in each resistance.

Solution

Given data

$$R_1 = 20 \Omega$$

$$R_2 = 30 \Omega$$

$$R_3 = 50 \Omega$$

1. Current flowing through each resistance

Current flowing through in resistance R_1 , $I_1 = \frac{V}{R_1}$

$$I_1 = \frac{60}{20}$$

$$I_1 = 3 \text{ Amp}$$

Current flowing through in resistance, R_2 , $I_2 = \frac{V}{R_2}$

$$I_2 = \frac{60}{30}$$

$$I_2 = 2 \text{ Amp}$$

Current flowing through in resistance, R_3 , $I_3 = \frac{V}{R_3}$

$$I_3 = \frac{60}{50}$$

$$I_3 = 1.2 \text{ Amp}$$

Total current in a circuit is $I = I_1 + I_2 + I_3$

$$I = 3 + 2 + 1.2$$

$$I = 6.2 \text{ Amp}$$

2. Total power

$$\text{Power } (P) = VI$$

$$P = 60 \times 6.2$$

$$P = 372 \text{ Watt}$$

3. Equivalent resistance

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{eq}} = \frac{1}{20} + \frac{1}{30} + \frac{1}{50}$$

$$\frac{1}{R_{eq}} = 0.05 + 0.033 + 0.02$$

$$\frac{1}{R_{eq}} = 0.103$$

$$R_{eq} = \frac{1}{0.103}$$

$$R_{eq} = 9.708 \Omega$$

4. Power loss in each resistance

$$\text{Power loss in resistance } R_1, P_1 = I_1^2 R_1$$

$$P_1 = 3^2 \times 20$$

$$P_1 = 9 \times 20$$

$$P_1 = 180 \text{ Watt}$$

$$\text{Power loss in resistance } R_2, P_2 = I_2^2 R_2$$

$$P_2 = 2^2 \times 30$$

$$P_2 = 4 \times 30$$

$$P_2 = 120 \text{ Watt}$$

$$\text{Power loss in resistance } R_3, P_3 = I_3^2 R_3$$

$$P_3 = 1.2^2 \times 50$$

$$P_3 = 1.44 \times 50$$

$$P_3 = 72 \text{ Watt}$$

$$\therefore \text{Total power loss } P_{is} = P_1 + P_2 + P_3$$

$$P = 180 + 120 + 72$$

$$P = 372 \text{ Watt}$$

Example 2

Two resistance R_1 and R_2 are connected in a parallel. If the current taken from The supply is 6 amperes, calculate the value of resistance R_1 , given that the value of R_2 is 6Ω and current through R_1 is 2 amperes. Also calculate the power.

Solution

$$\text{Current through in } 6\Omega \text{ resistance, } I = I_1 + I_2$$

$$I_2 = I - I_1$$

$$I_2 = 6 - 2$$

$$I_2 = 4 \text{ Amp}$$

$$\text{Voltage drop of the circuit } V = I_2 R_2$$

$$V = 4 \times 6$$

$$V = 24 \text{ Volt}$$

$$\text{Resistance } R_1 = \frac{V}{I_1}$$

$$R_1 = \frac{24}{2}$$

$$R_1 = 12 \Omega$$

$$\text{Power } P = V \times I$$

$$P = 24 \times 6$$

$$P = 144 \text{ Watt}$$

❖ UNSOLVE EXAMPLE

Example:1

Three resistance of 5, 12 and 20 Ω are connected in parallel. A resistance of 7 Ω is connected in series with this combination. The circuit is given 100 Volt supply. Calculate following 1) voltage drop across 7 Ω resistance. 2) Current flowing through 5 Ω resistance.

Example:2

A circuit consist of two parallel resistor having resistance of 20 Ω and 30 Ω respectively connected in series with 15 Ω resistance. If current through 15 Ω resistance is 3 amperes, find 1) current through 20 Ω and 30 Ω resistor. 2) The voltage across the whole circuit.

❖ Capacitor

Definition

The dielectric material placed between two conducting plates, capacitor is formed.

Capacitance

Property of capacitor to store charge is called capacitance.

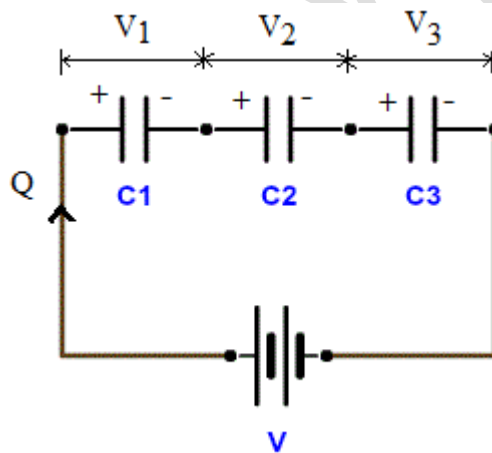
The unit of capacitor is Farad

$$\text{Charge } Q = CV$$

$$\therefore \text{Capacitance } C = \frac{Q}{V}$$

Capacitor in series connection

Diagram



- When the capacitor connected in series circuit the value of charge is same.
- Due to these current there is voltage drop across each capacitor.
- The voltage drop is proportional to the value of capacitor.
- Sum of voltage is equal to the applied voltage.

$$V = V_1 + V_2 + V_3$$

$$\therefore \frac{Q}{C} = \frac{Q_1}{C_1} + \frac{Q_2}{C_2} + \frac{Q_3}{C_3}$$

$$\therefore \frac{Q}{C} = Q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$$

$$\therefore \frac{1}{C} = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$$

$$\therefore \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

Example:

Capacitor of $10 \mu F$, $20 \mu F$ and $30 \mu F$ are connected in series and supply of 200 V d.c is given. Find the voltage across each capacitor

Given data:

$$C_1 = 10 \mu F$$

$$C_2 = 20 \mu F$$

$$C_3 = 30 \mu F$$

$$V = 200 \text{ Volt}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{C_{eq}} = \frac{1}{10} + \frac{1}{20} + \frac{1}{30}$$

$$\frac{1}{C_{eq}} = 0.1 + 0.05 + 0.0333$$

$$\frac{1}{C_{eq}} = 0.18333$$

$$C_{eq} = \frac{1}{0.18333}$$

$$C_{eq} = 5.45 \mu F$$

Charge

$$Q = CV$$

$$Q = 5.45 \times 10^{-6} \times 200$$

$$Q = 1.09 \times 10^{-3} \text{ OR } 0.00109$$

$$\text{Voltage drop across capacitor } C1, V1 = \frac{Q}{C1}$$

$$V1 = \frac{1.09 \times 10^{-3}}{10 \times 10^{-6}}$$

$$V1 = 109.1 \text{ Volt}$$

$$\text{Voltage drop across capacitor } C2, V2 = \frac{Q}{C2}$$

$$V1 = \frac{1.09 \times 10^{-3}}{20 \times 10^{-6}}$$

$$V1 = 54.5 \text{ Volt}$$

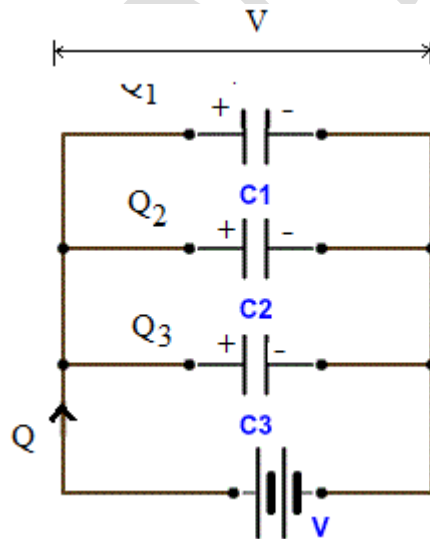
$$\text{Voltage drop across capacitor } C3, V3 = \frac{Q}{C3}$$

$$V1 = \frac{1.09 \times 10^{-3}}{30 \times 10^{-6}}$$

$$V1 = 36.3 \text{ Volt}$$

❖ Capacitor in parallel connection

Diagram



- The parallel connection of capacitor is shown in the diagram
- One lead of each capacitor are connected together and similarly other lead are connected together.
- Voltage 'V' applied across these two terminal. So each capacitor gets same voltage.
- The value of charge 'Q' is different and inversely proportional to each capacitor.

$$Q = Q_1 + Q_2 + Q_3$$

$$CV = C_1V + C_2V + C_3V$$

$$CV = V(C_1 + C_2 + C_3)$$

$$\therefore C = C_1 + C_2 + C_3$$

Example:

Capacitor of 1, 2, 4, and 6 μF are connected in parallel across 100 volt d.c supply. Find the equivalent capacitance and the charge on each capacitor.

1. Equivalent capacitor

$$C = C_1 + C_2 + C_3$$

$$C = 1 + 2 + 4 + 6$$

$$C = 13 \mu F$$

2. Charge

$$Q = CV$$

$$Q = 13 \times 10^{-6} \times 200$$

$$Q = 1.3 \times 10^{-3} \text{ Columb}$$

3. Charge across each capacitor

$$\text{charge across capacitor } C_1, Q_1 = CV$$

$$Q_1 = 1 \times 10^{-6} \times 100$$

$$Q_1 = 0.1 \times 10^{-3} \text{ Columb}$$

$$\text{charge across capacitor } C_2, Q_2 = CV$$

$$Q_2 = 2 \times 10^{-6} \times 100$$

$$Q_2 = 0.2 \times 10^{-3} \text{ Columb}$$

$$\text{charge across capacitor } C_3, Q_3 = CV$$

$$Q_3 = 4 \times 10^{-6} \times 100$$

$$Q_3 = 0.4 \times 10^{-3} \text{ Columb}$$

charge across capacitor $C4, Q4 = CV$

$$Q4 = 6 \times 10^{-6} \times 100$$

$$Q4 = 0.6 \times 10^{-3} \text{ Colum}$$

Active element

This is source of EMF or current. Source of EMF and current is known as an active element.

Passive element

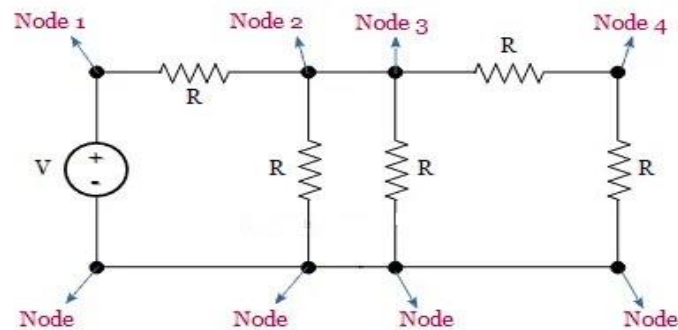
In D.C circuit passive element is the resistance only its known as a passive element.

Network

Circuit made by the connection of passive element or connection of active and passive elements is called the network.

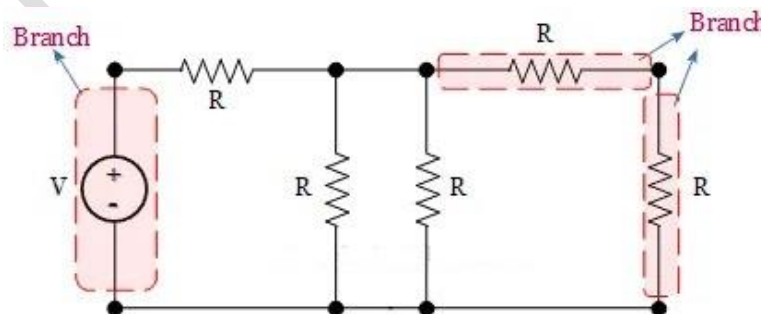
Node

The point at which two or more element meet is called the node.



Branch

Element or group of elements connected between two nodes is called branch.



- In a diagram, one node **H** of a network is shown. In this I_1 , I_4 and I_5 flowing towards the node so these are assigned positive sign
- whereas current I_2 , I_3 and I_6 are flowing away from the node so these assigned negative sign.

$$\therefore I_1 - I_2 - I_3 + I_4 + I_5 + I_6 = 0$$

$$\therefore I_1 + I_4 + I_5 = I_2 + I_3 + I_6$$

Sum of current flowing toward node = sum of currents flowing away from the node.

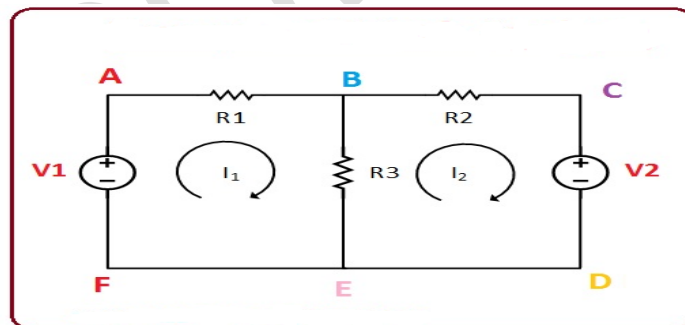
2. Kirchhoff's voltage law

- The algebraic sum of voltages in a closed loop is zero.

$$\Sigma V = 0$$

This may be voltage drop or voltage rise.

- For example, current flow through the element from higher potential to lower potential, the volt is drop. If current assumed to flow from the positive terminal of battery to the negative terminal of battery it is voltage drop.
- But if the current is assumed to flow from negative terminal of the battery to the positive terminal, it is voltage rise.
- Voltage drop taken as negative and voltage rise is taken as positive
- In any closed loop of a network, Sum of voltage drops = Sum of voltage rise



❖ Explain Active and Passive component with Example

Active element

This is source of EMF (Voltage) or current. Source of EMF and current is known as an active element.

Voltage source

There is no decrease in the voltage of the voltage source when any amount of current is drawn from the voltage source.

- This means that the internal resistance of the ideal voltage source is zero.
- But ideal voltage source is not possible in actual practice.
- So when current drawn from the voltage source some voltage drop occurs due to internal resistance so terminal voltage of the voltage source decrease.

Current source

Current source supplies the current. Current source can also be ideal or practical.

- The ideal current source can supply constant current to any value of the resistance connected across it.
- The ideal current source has no resistance

Passive component

The component which cannot deliver power or cannot process the electrical signal are known as the passive component.

There are three types passive component.

- Resistance
- Inductor
- Capacitor

1. Resistance

The property of material to oppose the flow of electric current through it is known as the resistance.

2. Inductor

An inductor consists of a coil or wire. if an iron core is inserted in it, it is called an iron core inductor.

When core is not used it is called the air core inductor.

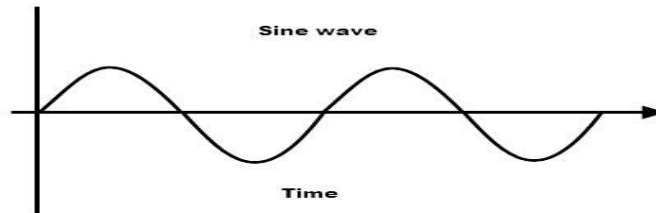
3. Capacitor

When insulating material placed between two conducting plates a capacitor is formed.

❖ Important term related to alternate quantity:

1. Sinusoidal signal

In sine wave signal the magnitude of the signal varies as the sign of the angular frequency.



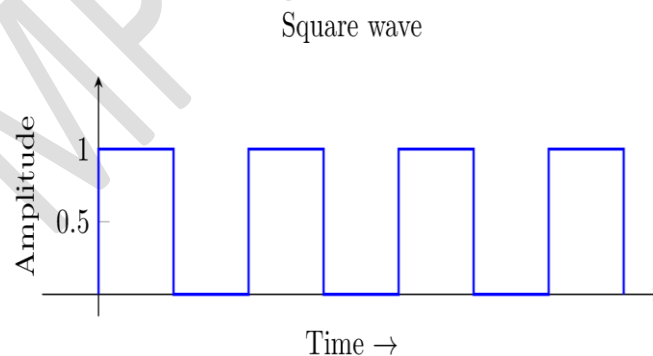
2. Triangular wave

A triangle wave is non-sinusoidal waveform named for its triangular shape.



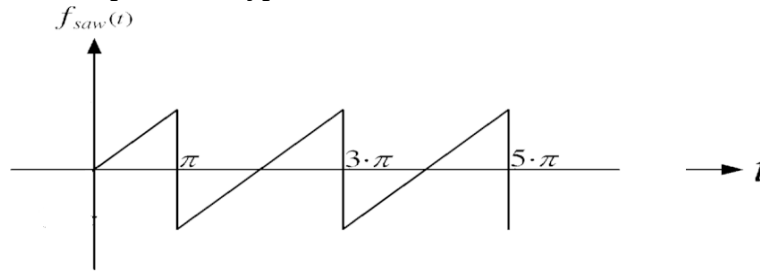
3. Square wave

A square wave is a periodic wave that alternates between two fixed amplitudes for equal length of time. This signal is of symmetrical type or of asymmetrical type.



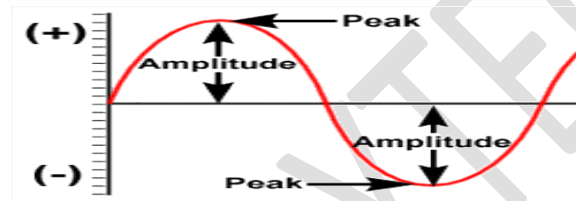
4. Saw tooth wave

The shape of this type of wave is like the tooth of the saw.



5. Amplitude

Amplitude is the maximum value of current or voltage.



6. Frequency

The numbers of cycle of completed in one second is known as the frequency.

The unit of frequency is Hertz.

It is denoted by 'f'

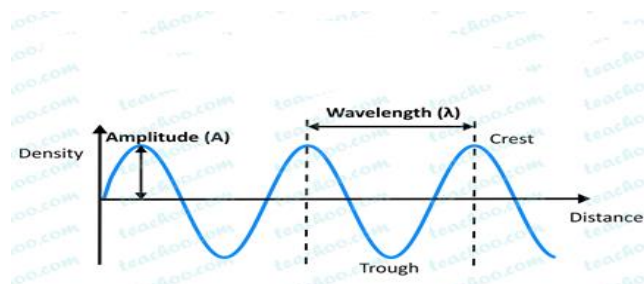
7. Wave length

Wavelength means the length of one wave.

The distance between two consecutive positive peaks A and B is called the wavelength.

The unit of wavelength is Meter

It is denoted by ' λ '



8. Time period

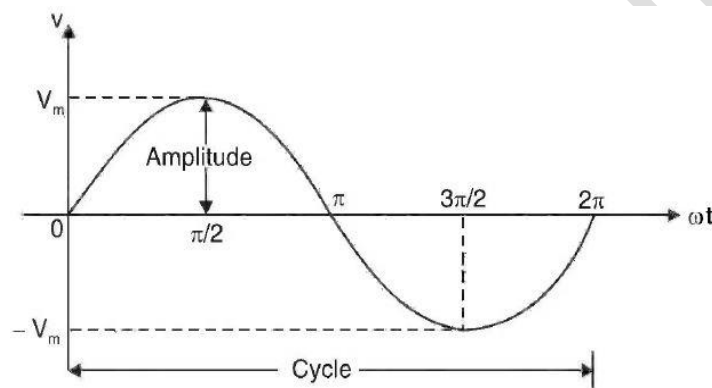
The time taken to complete one cycle is known as the time period

The unit of time period is Second

It is denoted by the 'T'

9. Cycle

The value of EMF increase in one direction from zero becomes the maximum and the decrease and become zero. This complete alteration is known as the cycle.



10. Instantaneous value

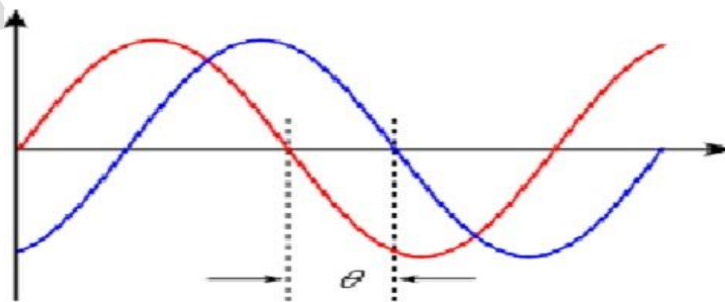
Value of EMF or current at any time is called instantaneous value.

11. Phase

In alternating quantity, phase means the time required to reach certain value is called phase.

12. Phase difference

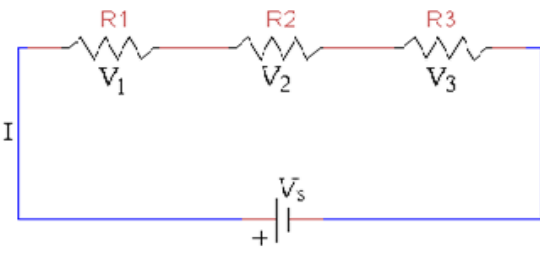
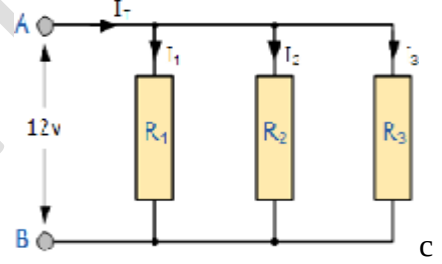
When two alternating quantities become maximum in same direction, the difference in phase of the maximum values is called phase difference.



❖ Give the difference between EMF and potential difference

Sr. No.	EMF (Electro motive force)	Potential Difference
1.	The EMF means the force require for the flow of electric current	Potential difference means the potential difference between two charge body
2.	The EMF of a cell is the maximum potential difference between two electrodes of a cell, when the cell is in the open circuit.	The terminal potential difference of a cell is the potential difference between the two terminal of the cell in a close circuit.
3.	It is independent of the resistance of the circuit.	It is depends upon the resistance of the circuit and current flowing through in it.
4.	The term EMF is used for the source of electric current	The potential difference is measured between any two point of the electric circuit
5.	The EMF is a cause	The potential difference is an effect

❖ Give the difference between EMF and potential difference

Sr. No.	Series connection	Parallel connection
1		
2	The current passing through all the elements connected in series is the same	Voltage across each elements connected in parallel is the same
3	There is a only one path for flow of current	The path for flow of current is more than one.
4	The total voltage applied across series circuit is equal to the sum of the voltage drop across all the elements connected in series i.e. $V = V_1 + V_2 + V_3$	The total current flowing through Parallel combination is equal to the sum of all current flowing through all the elements connected in parallel. i.e. $I = I_1 + I_2 + I_3$
5	The equivalent resistance is $R_{eq} = R_1 + R_2 + R_3$	The equivalent resistance is $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
6	The equivalent resistance is greater than the greatest resistance connected in series.	The equivalent resistance is less than that of the least resistance connected in the parallel circuit.